CURRICULUM VITAE

Personal Information

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Current position	Visiting Researcher (100%) - CAIR, UiA
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Professional Summary

Research oriented **AI Engineer and Visiting Researcher at CAIR with 2+ years of experience in** interpretable AI, generative models, autonomous sensing systems, and machine learning for real-world environments.

My recent work focuses on interpretable generative AI using Tsetlin Machines, probabilistic modeling ,and distributed intelligent systems. I am particularly interested in AI safety, alignment, and trustworthy autonomous agents, long-term reasoning systems, and evaluation of adaptive AI behaviour in real-world settings.

I enjoy working on foundational yet practical research problems that improve reliability, transparency, and controllability of advanced AI systems.

Research Interests:

- Autonomous AI Agents
- Autonomous Systems
- Trustworthy Machine Learning
- Computer Vision
- AI safety and Alignment
- Multi-Agent Systems
- Generation AI and LLM Evaluation
- Human-AI Interaction and Cooperative AI
- Context-Aware Intelligent Systems.

Education

08. 2023 – 05.2025 **Master's in Computer Science – AI & Data Science**
IIITDM Kancheepuram, Chennai, India
Exchange Student at UiA: University of Agder, Norway (from 2024-2025)
Grade: A
Master's thesis: "Distributed FMCW Radar Point Cloud Clustering and Fusion Framework for Human Intruder Application"

- I designed and tested a complete radar point cloud clustering, tracking, and fusion framework for detecting and following multiple human intruders. My contributions were:
  (1) A self-tunable DBSCAN for sparse/non-uniform radar data,
  (2) A multi-radar system with dual-mode clustering and distributed fusion using EKF-GNN tracking and SCI for global trajectory

estimation,

(3) An ensemble spectral clustering pipeline with consensus-based DBSCAN variants, validated through real-world deployments on AWR1843 radars with Raspberry Pi nodes.

08.2019 – 06.2023 **Bachelor's in Information Technology**

JNTUH College of Engineering, Hyderabad, India

CGPA: 7.63 (equivalent to Grade C)

Bachelor's thesis: "Real-Time Image Segmentation for Self Driving Cars"

- I designed and implemented a deep learning-based semantic segmentation system tailored for self-driving cars, focusing on real-time scene understanding and object boundary detection to enhance autonomous navigation accuracy.

Research Experience

01.2025– till date **Research Assistant – CAIR**, UiA, Norway

- Researching interpretable generative AI using Tsetlin Machines for probabilistic image modeling
- Develop and implement models using Python

Aug 2024 - May 2025 **Visiting Researcher**, UiA, Norway

In this position, the main responsibilities were:

- Project 1: Human Activity Recognition using CSI signals (**Outcome:** APSCON 2025).
- Project 2: Multi-Channel 1D-CNN Architecture for Drone vs Bird Classification Using mmWave Radar ADC Data. (**Outcome:** INDICON 2025)
- Project 3: Machine Learning and Deep Temporal Networks for UAV Velocity Classification Using High Dimensional CSI and RSSI Data. (**Outcome:** IJAIA 2026)

Key Competences

- Strong background in Artificial Intelligence and Machine Learning, with experience developing data-driven systems and generative AI applications.
- Proficient in Python and modern ML frameworks, with hands-on experience implementing, evaluating, and optimizing models in research and production-oriented workflows.
- Experience working with structured and unstructured data, including feature engineering, data preprocessing, and building scalable pipelines for analytics and AI applications.
- Familiar with LLM-based systems, including RAG architectures, embedding models, and vector search, with an interest in AI agents and knowledge retrieval systems.
- Knowledge of deep learning for computer vision and sequential modeling, along with exposure to NLP and document understanding tasks.
- Strong foundation in interpretable AI, including research experience with Tsetlin Machines, for transparent and efficient modeling.
- Experience conducting reproducible machine learning experiments, model evaluation, and performance optimization, with a focus on reliability, scalability, and robustness.
- Strong analytical and research skills, with ability to translate complex data and experimental findings into actionable insights for real-world applications.

Languages

- English – Fluent in written and oral

Tools, Technologies and Software Skills

Programming & Data Engineering

- Python, SQL, Bash
- PySpark, Data Pipelines, ETL Processing

Generative AI & LLM Systems

- LLMs, Prompt Engineering, RAG Architecture
- LangChain, Hugging Face Transformers
- Embedding Models, Vector Search, Semantic Retrieval
- AI Agents (LangGraph, OpenAI Agents SDK – familiarity)

Machine Learning & AI

- Deep Learning: CNNs, Transformers, Representation Learning
- Sequential Modeling, Time-Series Learning, Classification Systems
- Model Evaluation, Optimization, Reproducible Experimentation
- Computer Vision, NLP, Semantic Search, Document Understanding

Interpretable & Trustworthy AI

- Explainable AI (SHAP, LIME)
- Tsetlin Machines and Rule-Based Interpretable Learning
- Reliable and Transparent AI Modeling

Autonomous & Distributed Intelligent Systems

- Multi-Sensor Fusion and Distributed Learning Systems
- Autonomous Perception and Tracking Systems
- Real-world AI Deployment and Edge AI Applications

MLOps & Infrastructure

- Docker, Kubernetes
- GitHub Actions, Jenkins
- Model Deployment, Monitoring, Experiment Tracking
- Reproducible ML Pipelines

Cloud & Infrastructure

- AWS (EC2, S3, IAM),
- Infrastructure as Code (Terraform)
- Linux, Networking, Distributed Systems Basics

Courses

- "Beginners, Zero to Hero. AWS EC2 web server, NodeJS Server, AWS RDS database server, S3, SES & CloudWatch" by BackSpace Academy – UDEMY
- "100 days of code: The complete python pro bootcamp" by Dr. Angela Yu – UDEMY
- "Mathematics - Basic to Advanced for Data Science and GenAI" by Krish Naik – UDEMY
- "Complete Generative AI course with Langchain and Huggingface" by Krish Naik – UDEMY

Academic Articles

08.2024 – 05.2025

- "Lightweight 2D CNN Fusion for Human Activity Recognition Using Multi-Transceiver CSI Data on Edge Devices" – **APSCON 2025**
- "Multi-Channel 1D-CNN Architecture for Drone vs Bird Classification Using mmWave Radar ADC Data." – **INDICON 2025**
- "Machine Learning and Deep Temporal Networks for UAV Velocity Classification Using High Dimensional CSI and RSSI Data." – **IJAIA 2026**

- “Corrective Loss: Penalising Confident Misclassifications via False-Class Probability Suppression ” – **BMVC 2026** (**Under Review**)
- **IEEE Transactions on Radars (Under Review)**
  - ”Space-Time Adaptive DBSCAN for Multi-Human Intruder Tracking using Automotive Radar”
  - ”Dual-Mode Clustering and Fusion for Intruder Tracking with Distributed FMCW Radars”
  - ”Spectral Clustering & Multi-Sensor Fusion for Radar Intruder Tracking”

REFERENCES

- **Linga Reddy Cenkeramaddi (Master’s thesis supervisor at UiA)**

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